OceanEye-AI Vision-driven Underwater Robot Intelligent Biological Detection System

【学生 1：水下机器人操作与数据采集】

Our project is called:
“OceanEye — AI Vision-driven Underwater Robot Intelligent Biological Detection System.”

With the rapid development of artificial intelligence and marine technology, underwater intelligent perception has become increasingly important in marine monitoring, ecological protection, and underwater exploration.

However, underwater environments are very complex. Due to light absorption and scattering, underwater images usually suffer from color distortion, low contrast, blurred edges, and unclear biological targets.

Therefore, our project combines an underwater robot platform with artificial intelligence technology to achieve intelligent underwater biological detection.

This is our underwater robot platform.

The robot integrates underwater cameras, motion control modules, communication modules, and image acquisition systems.

During operation, the robot can move underwater and collect real-time visual information.

The complete workflow is:
Underwater robot image acquisition → AI image processing → Target detection → Result analysis

Through this process, the system can automatically discover and recognize underwater biological targets.

Next, my teammate will demonstrate how our AI algorithm processes underwater images and performs target detection.

【学生 2：AI 视觉算法运行与目标检测】

Hello everyone.

I will introduce the artificial intelligence part of our system.

The main challenge of underwater biological detection is that targets usually have small sizes, similar backgrounds, unclear boundaries, and complex underwater environments.

For example, sea cucumbers, sea urchins, scallops and starfish are often difficult to identify.

To solve these problems, we use a deep learning-based object detection algorithm.

This is the original underwater image collected by the robot.

After AI processing, the system can automatically detect biological targets.

The detection results include target category, confidence score, and location information.

Our detection algorithm is based on an improved YOLOv8 model.

We introduce attention enhancement, global feature modeling and multi-scale feature fusion technologies.

These improvements allow the system to achieve more accurate and stable detection results in complex underwater environments.

When the underwater robot moves, the AI system can continuously analyze the video stream and detect biological targets automatically.

【学生 3：检测结果分析与应用展示】

Hello everyone.

I will introduce the detection results and application value of our system.

After processing by the AI detection model, the system can obtain detailed information about underwater biological targets.

The output includes object category, target location, and detection confidence.

Compared with traditional underwater inspection methods, our intelligent system improves detection efficiency and reduces manual observation workload.

Our system can be applied in marine ecological monitoring, underwater robot inspection, and intelligent marine platforms.

The robot can identify underwater organisms, support marine resource investigation, and assist underwater exploration.

In summary, our project integrates underwater robots, computer vision, and artificial intelligence algorithms.

It realizes the complete process from underwater image acquisition to intelligent biological target recognition.

Through AI technology, we hope to make underwater exploration more intelligent, efficient, and reliable.

Thank you for your attention.